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SmartHome Gesture Control Application Project Part 2 Report

Approach

I approached the problem by organizing my training videos and testing videos into separate folders as well as a folder for train frames and test frames. I loaded the CNN model then used the frame extractor to get the middle frame from each video. I looped through the training and testing video folders and saved each extracted middle frame into its respective frames folder. Then I loaded each frame image and passed it through the extractor to obtain its feature vector. I stored the training and testing feature vectors in separate lists.

Solution

I combined the training feature vectors into a matrix so each test vector can be compared against all training vectors by using the cosine distance function. For each test vector, I selected the training vector with the minimum cosine distance as the closest match. Then I mapped each gesture to its output label and saved the resulting predictions to Results.csv.